

NASHVILLE INTL, TN, USA

WMO: 723270

Lat: 36.119N

Lon: 86.689W

Elev: 600

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	19.5	1.2	6.0	21.3	6.0	7.6	22.7	24.0	53.0	20.5	53.2	6.8	330	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	94.4	74.7	92.3	74.5	90.4	73.9	78.1	88.8	77.1	87.5	76.1	85.9	6.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	134.8	82.8	74.0	129.6	81.9	73.0	125.4	81.1	41.8	89.4	40.7	87.5	39.8	86.2	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	17.0	15.3	DB	9.1	97.6	5.4	3.5	5.2	100.2	2.0	102.2	-1.0	104.2	-5.0	106.7
			WB	7.5	79.8	5.0	1.6	3.9	81.0	0.9	82.0	-1.9	82.9	-5.6	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	38.9	43.0	51.0	60.5	68.9	76.7	79.9	79.3	72.9	61.5	50.1	42.5
	DBStd	16.69	11.70	10.90	10.01	8.56	7.07	4.83	3.85	4.17	6.39	8.24	9.05	10.16
	HDD50	1149	379	246	112	16	0	0	0	0	9	109	278	
	HDD65	3430	810	617	445	183	46	1	0	13	166	451	698	
	CDD50	4994	35	50	142	331	587	800	927	910	686	366	113	47
	CDD65	1800	0	1	9	49	168	352	462	445	250	58	5	1
	CDH74	17422	0	7	84	438	1478	3411	4765	4474	2229	502	33	1
	CDH80	7014	0	0	7	77	458	1407	2100	1984	860	118	3	0
	WSAvg	6.7	7.8	8.0	8.2	7.9	6.5	5.7	5.3	5.1	5.4	6.0	6.8	7.4
	PrecAvg	50.9	5.0	4.3	5.0	4.4	5.0	3.9	4.4	3.5	3.5	3.2	3.6	5.1
Precipitation	PrecMax	60.8	9.8	10.0	10.6	7.9	15.6	10.7	8.3	8.5	8.7	8.2	6.2	12.3
	PrecMin	38.5	2.1	2.0	2.8	0.9	0.5	0.1	2.0	1.4	0.2	0.0	0.6	2.4
	PrecStd	5.7	1.6	1.9	1.9	2.0	2.8	2.2	1.7	1.8	2.0	1.9	1.5	2.3
	PrecStd	5.7	1.6	1.9	1.9	2.0	2.8	2.2	1.7	1.8	2.0	1.9	1.5	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	69.4	74.3	80.7	85.4	90.7	94.7	97.2	98.5	94.9	87.6	78.5	71.8	
	0.4% MCWB	60.7	61.9	63.5	66.4	71.9	73.8	76.2	73.0	72.3	70.2	64.2	61.9	
	2% DB	65.5	69.0	76.2	82.3	87.5	92.1	93.7	94.7	91.3	83.1	72.8	66.9	
	2% MCWB	58.8	58.5	60.9	64.6	70.0	74.2	76.0	74.6	71.8	67.6	60.8	60.1	
	5% DB	62.0	64.8	72.1	79.2	85.1	90.0	91.8	92.0	88.5	79.6	69.3	62.5	
	5% MCWB	56.5	55.8	58.4	63.2	68.9	73.5	75.5	74.5	71.2	65.1	59.2	56.9	
	10% DB	57.6	60.6	67.9	75.9	82.4	87.8	89.7	89.4	85.3	76.0	65.4	58.9	
	10% MCWB	51.4	52.5	55.9	62.0	68.3	72.3	74.6	73.8	70.0	64.1	57.2	53.6	
	0.4% WB	63.7	64.4	65.9	70.3	75.2	78.2	79.6	80.2	76.1	73.3	67.0	64.9	
	0.4% MCDB	67.0	71.0	76.1	80.8	84.2	89.1	90.4	90.1	86.8	82.0	73.2	69.8	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	60.3	61.6	63.3	68.1	73.0	76.8	78.3	78.0	74.7	70.9	64.2	61.6	
	2% MCDB	64.3	67.9	72.6	77.6	83.1	87.8	89.2	88.1	84.6	78.6	70.6	65.9	
	5% WB	57.4	58.0	60.9	66.0	71.4	75.6	77.3	76.6	73.4	68.7	61.5	58.1	
	5% MCDB	62.3	62.5	69.3	74.9	81.4	85.9	88.2	86.3	82.6	75.4	67.4	61.5	
	10% WB	52.5	54.0	58.4	64.1	70.0	74.5	76.3	75.5	72.1	66.5	58.6	54.0	
	10% MCDB	56.4	59.1	66.2	72.6	79.1	84.1	86.3	84.8	81.0	73.5	63.9	58.5	
Mean Daily Temperature Range	MDBR	17.0	18.3	20.3	21.3	19.7	18.8	18.0	18.6	20.1	21.1	19.8	16.6	
	5% DB MCDBR	20.3	23.3	25.6	24.7	22.9	21.5	20.2	21.6	23.2	24.9	24.3	20.1	
	5% DB MCWBR	15.0	15.7	14.1	11.4	9.3	7.4	6.5	6.8	7.7	11.6	14.4	14.9	
	5% WB MCDBR	18.1	19.2	21.7	19.7	19.2	18.8	18.2	17.9	18.6	18.6	18.8	17.4	
	5% WB MCWBR	14.8	15.4	14.1	11.0	9.1	7.7	6.9	6.9	7.6	10.3	13.7	14.7	
	5% WB MCWBR	14.8	15.4	14.1	11.0	9.1	7.7	6.9	6.9	7.6	10.3	13.7	14.7	
Clear-Sky Solar Irradiance	taub	0.319	0.326	0.373	0.402	0.452	0.477	0.509	0.521	0.430	0.374	0.335	0.311	
	taud	2.478	2.449	2.347	2.265	2.172	2.138	2.067	2.021	2.261	2.404	2.467	2.514	
	Ebn at Noon	279	290	283	279	266	259	249	243	261	268	270	274	
	Edh at Noon	27	30	37	43	47	49	52	54	40	32	27	24	
All-Sky Solar Radiation	RadAvg	702	882	1235	1617	1810	1982	1889	1742	1483	1151	827	602	
	RadStd	60	110	113	111	150	133	102	84	124	142	86	58	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+272	+182	
(47)	Regional (19 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+176	

Nomenclature: See separate page